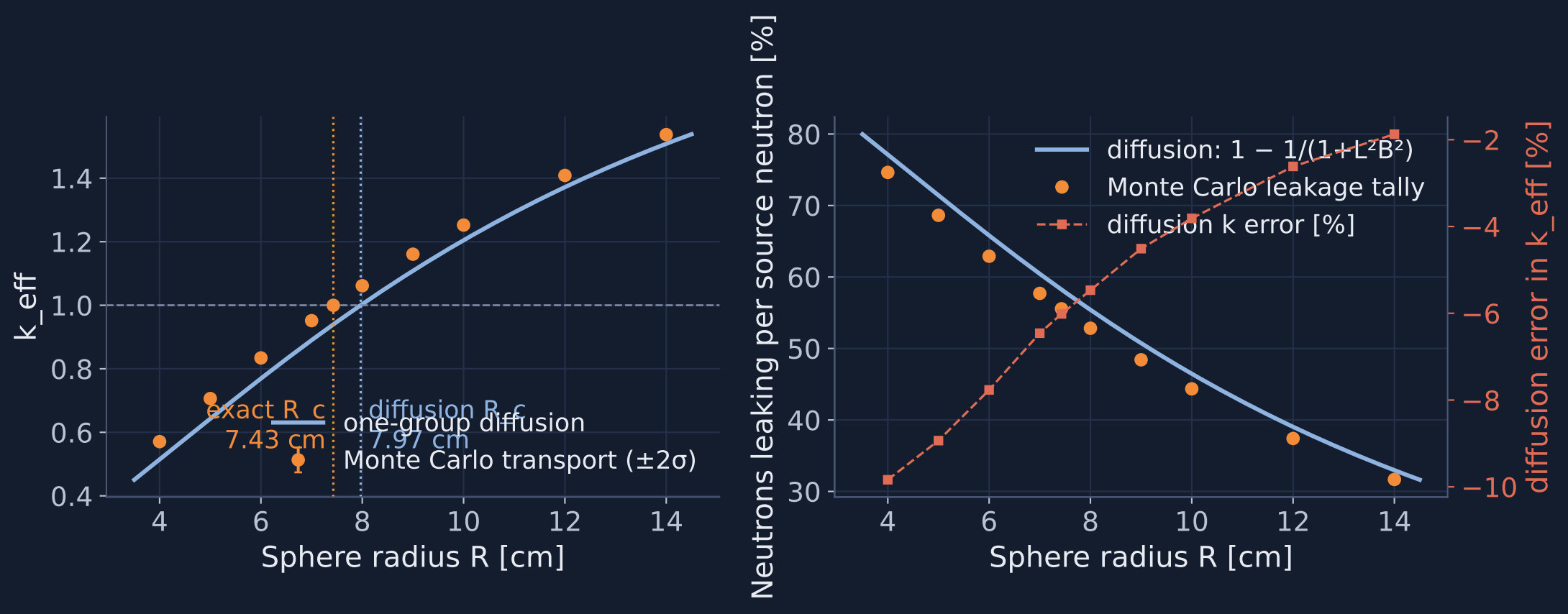
k_eff
1.4
1.2
1.0
0.8
0.6
0.4
4
6
8
10
12
14
Sphere radius R [cm]
exact R_c
7.43 cm
diffusion R_c
7.97 cm
one-group diffusion
Monte Carlo transport (±2σ)
Neutrons leaking per source neutron [%]
80
70
60
50
40
30
diffusion error in k_eff [%]
−2
−4
−6
−8
−10
diffusion: 1 − 1/(1+L²B²)
Monte Carlo leakage tally
diffusion k error [%]